

# Seohee Choy

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## EDUCATION

### University of Wisconsin - Madison

Bachelor of Science in Computer Science and Data Science | GPA: 3.88 / 4.0

Madison, WI

September 2023 - May 2027

## RESEARCH / EXPERIENCE

### Wisconsin Undergraduate Research in Data Science (WISCURDS)

Madison, WI

Undergraduate Researcher

July 2026 - Present

- Develop an **LLM-powered dual-agent** system in collaboration with the **Data Science Institute** and **Wisconsin Evaluation Collaborative** to automate **WIDA** policy research by recursively crawling content from seed URLs.

### Robust Intelligence and Robotics Lab @ KAIST | [Lab Website](#)

Daejeon, South Korea

Undergraduate Researcher, Visiting Scholar (Advisor: Professor DaeHyung Park)

November 2025 - Present

- Conduct research on active multi-view visual inputs to improve manipulation robustness under occlusion.
- Design and manufacture remote gimbal camera modules using **Fusion 360, Raspberry Pi, BLDC motors, and FOC drivers** to improve visual access and reduce task-space occlusions in a **UR5e** workspace.
- Develop **Python/PyTorch-based manipulation policies** in **Isaac Sim**, experimenting with **HIL-SERL, PPO, and ACT**, and integrate SpaceMouse teleoperation for human demonstrations and intervention.

### Vision Lab @ LG AI Research | [LG AI Talk Concert](#)

Seoul, South Korea

AI Engineer Intern

May 2025 - October 2025

- Developed crawling pipelines using **CLIP embeddings** to curate large-scale multimodal QA datasets across **DocVQA, ChartQA, MMMU, and AI2D** for training **EXAONE-VL**.
- Processed **300K+** multimodal samples into **vision-language model (VLM)**-compatible formats with automated quality control for modality completeness, image availability, and formatting constraints using **Qwen 2.5-VL**.
- Built a full-stack data labeling and validation platform for **20+ team members** using **TypeScript, Node.js, FastAPI, and PostgreSQL**. Restricted access to the company's internal domain to protect confidential data.

### University of Wisconsin - Madison, Department of Computer Science

Madison, WI

Peer Mentor for CS 540 (Introduction to AI)

August 2024 - December 2024

- Mentored students on **Python-based AI/ML projects** covering search, neural networks, and computer vision.
- Provided 6 hours of weekly 1:1 support to **10+ students** by translating complex algorithmic concepts into practical implementation and debuggable strategies.

## PROJECTS

### FIRST Tech Challenge Korea (Branksome Hall Asia Robotics Academy)

December 2020 - May 2023

- Led a **25 member team** across mechanical engineering, software programming, and logistics.
- Developed **Java-based vision recognition and autonomous navigation logic** via live webcam input.

### FarmBot (Smart Farming System) | [Media](#)

August 2020 - April 2023

- Co-led the hardware assembly, deployment, and sensor calibration of the **CNC-based FarmBot system**.

### Full-Stack Study Room Reservation System | [Demo](#)

May 2022 - May 2023

- Built a **TypeScript, React, and Firebase** reservation platform for **400+** students, implementing real-time validation, duplicate booking prevention, and synchronized reservation management.

## AWARDS

Dean's List, University of Wisconsin-Madison

Fall 2023, Spring 2024

The Wisconsin Alumni Association Korea Chapter (WAAK) Scholarship

Fall 2023

## TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, SQL, HTML/CSS

AI/ML & Frameworks: PyTorch, TensorFlow, Isaac Sim, OpenCV, React, React Native, FastAPI, Node.js

Tools & Platforms: Linux, Git, Fusion 360, Docker, Google Cloud Platform (GCP), Firebase